Attention Is All You Need: Empirical Study

Authors: Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose the Transformer, a novel network architecture based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results by over 2 BLEU.

1. Introduction & Methodology

Recurrent neural networks, particularly long short-term memory (LSTM) and gated recurrent (GRU) neural networks, have been firmly established as state of the art approaches in sequence modeling. To allow building representations of sequences without regard to their distance in input or output sequences, we propose Multi-Head Self-Attention. Multi-Head Attention allows the model to jointly attend to information from different representation subspaces at different positions.

2. Experimental Results & Benchmarks

On the WMT 2014 English-to-German translation task, the Transformer (big) achieves 28.4 BLEU score outperforming all previously reported models. On the WMT 2014 English-to-French translation task, our model achieves a new state-of-the-art BLEU score of 41.8 after training for 3.5 days on 8 P100 GPUs.

Model	WMT 2014 En-De (BLEU)	WMT 2014 En-Fr (BLEU)	Training Cost (FLOPs)
ByteNet	23.75	-	1.0e19
ConvS2S	25.16	40.46	9.6e18
Transformer (base)	27.3	38.1	3.3e18
Transformer (big)	28.4	41.8	2.3e19

3. Limitations & Conclusion

While the Transformer achieves state-of-the-art performance, it exhibits quadratic computational memory complexity O(N^2) relative to sequence length. In conclusion, we presented the Transformer, the first sequence transduction model based entirely on attention.